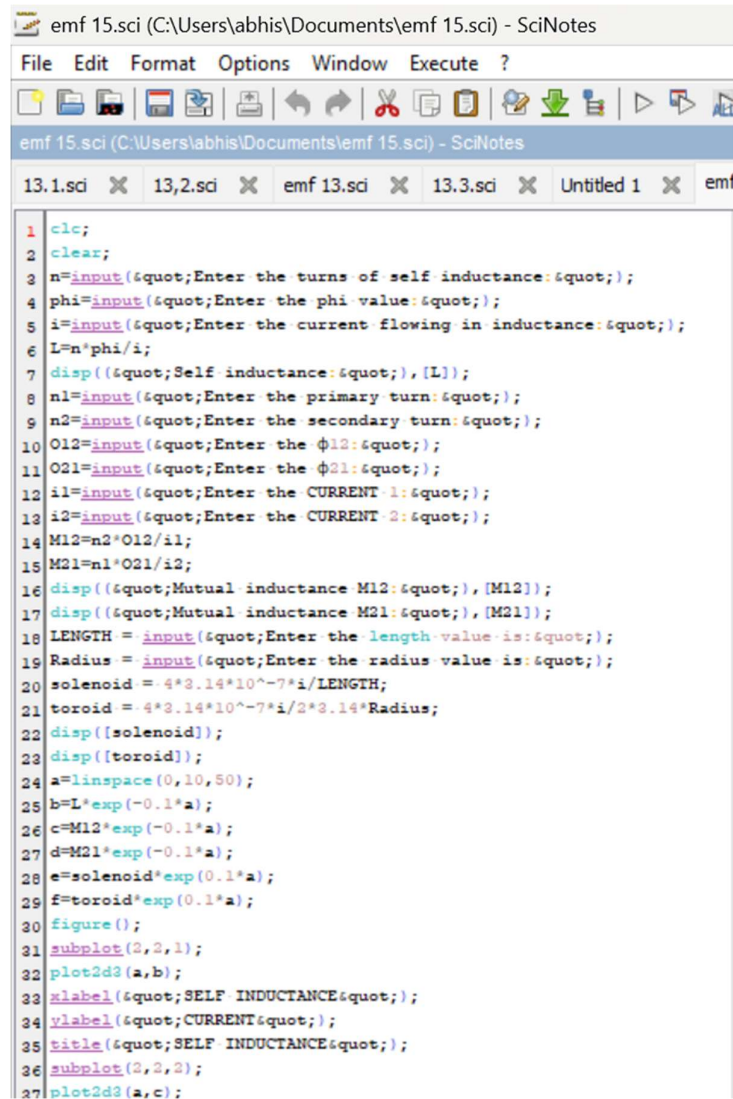


Exp No:15 Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Aim: To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1



```
1 clc;
2 clear;
3 n=input('Enter the turns of self inductance:');
4 phi=input('Enter the phi value:');
5 i=input('Enter the current flowing in inductance:');
6 L=n*phi/i;
7 disp('Self inductance:',[L]);
8 n1=input('Enter the primary turn:');
9 n2=input('Enter the secondary turn:');
10 O12=input('Enter the phi12:');
11 O21=input('Enter the phi21:');
12 i1=input('Enter the CURRENT 1:');
13 i2=input('Enter the CURRENT 2:');
14 M12=n2*O12/i1;
15 M21=n1*O21/i2;
16 disp('Mutual inductance M12:',[M12]);
17 disp('Mutual inductance M21:',[M21]);
18 LENGTH = input('Enter the length-value is:');
19 Radius = input('Enter the radius value is:');
20 solenoid = 4*3.14*10^-7*i/LENGTH;
21 toroid = 4*3.14*10^-7*i/2*3.14*Radius;
22 disp([solenoid]);
23 disp([toroid]);
24 a=linspace(0,10,50);
25 b=L*exp(-0.1*a);
26 c=M12*exp(-0.1*a);
27 d=M21*exp(-0.1*a);
28 e=solenoid*exp(0.1*a);
29 f=toroid*exp(0.1*a);
30 figure();
31 subplot(2,2,1);
32 plot2d3(a,b);
33 xlabel('SELF INDUCTANCE');
34 ylabel('CURRENT');
35 title('SELF INDUCTANCE');
36 subplot(2,2,2);
37 plot2d3(a,c);
```

Scilab 2026.1.0 Console

File Edit Control Applications ?



Scilab 2026.1.0 Console

Enter the turns of self inductance:200

Enter the phi value:3.14

Enter the current flowing in inductance:2

```
"Self inductance:"  
314.
```

Enter the primary turn:100

Enter the secondary turn:300

Enter the $\phi_{12}:5 \times 10^{-3}$

Enter the $\phi_{21}:6 \times 10^{-3}$

Enter the CURRENT 1:3

Enter the CURRENT 2:4

```
"Mutual inductance M12:"  
0.5
```

```
"Mutual inductance M21:"  
0.15
```

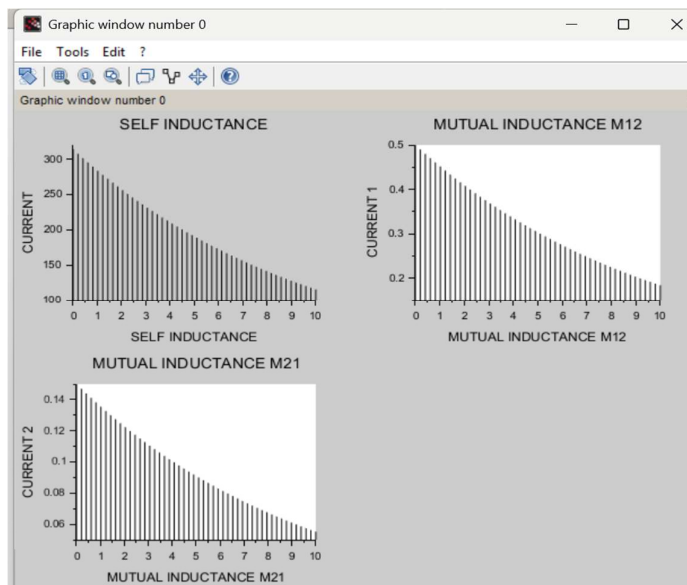
Enter the length value is:6

Enter the radius value is:2

0.0000004

0.0000079

exec: Wrong number of output argument(s): 0 expected.



Exp 15

$$N = 200$$

$$i = 2$$

$$n_1 = 100$$

$$n_2 = 300$$

$$\text{length} = 6 \text{ cm}$$

$$i_1 = 3 \text{ A}$$

$$i_2 = 4 \text{ A}$$

$$\Phi_{12} = 5 \times 10^{-3}$$

$$\Phi_{21} = 6 \times 10^{-3}$$

$$\text{Radius} = 2 \text{ cm}$$

Determine (i) self induction and MI (ii) solenoid and toroid with different length and cross sectional area.

$$N = 200, \Phi = 3.14 \text{ Wb}, I = 2 \text{ A}$$

$$L = \frac{N\Phi}{I}$$

$$L = \frac{200(3.14)}{2}$$

$$L = 314$$

(i) $N_2 = 300, \Phi_{12} = 5 \times 10^{-3}, I_1 = 3$

$$M_{12} = \frac{N_2 \Phi_{12}}{I_1}$$

$$M_{12} = \frac{300(5 \times 10^{-3})}{3}$$

$$M_{12} = 0.5 \text{ H}$$

(ii)

$$M_{21} = \frac{N_1 \Phi_{21}}{I_2}$$

$$M_{21} = \frac{100(6 \times 10^{-3})}{4}$$

$$M_{21} = 0.15 \text{ H}$$

Length = 6, Radius = 2

Solenoid turns. $N = 200$

Length $l = 6m$

Radius $r = 2m$

$$A = \pi r^2 = \pi (2)^2 = 12.566 m^2$$

$$L_{\text{solenoid}} = \frac{\mu_0 N^2 A}{l}$$

$$L_{\text{solenoid}} = \frac{(4\pi \times 10^{-7}) (200)^2 (12.566)}{6}$$

$$L_{\text{solenoid}} = 2.105 \times 10^{-2} H$$

0.0000004 H

0.0000079 //

Result: Thus determination and plotting of Inductance, Self-inductance, Mutual inductance of current, carrying coil, solenoid and toroid with different (a) length and (b) circumferential area Are performed and verified successfully.